

Lecture Notes: Functions and Sets

Calculus I

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Section 1 Sets and Intervals

Definition (Set). A *set* is a collection of distinct objects, considered as an object in its own right. The objects in a set are called its *elements*. If a is an element of set A , we write $a \in A$. If a is not an element of A , we write $a \notin A$. The set containing no elements is called the *empty set* and is denoted by $\emptyset$. **Definition (Subset).** If every element of set A is also an element of set B , then A is a *subset* of B , denoted $A \subseteq B$. If $A \subseteq B$ and $B \subseteq A$, then $A = B$. If there exists at least one element in B that is not in A , then A is a *proper subset* of B , denoted $A \subset B$. **Definition (Set Operations).** Let A and B be sets.

- The *union* of A and B , denoted $A \cup B$, is the set of all elements that are in A or in B (or both).
- The *intersection* of A and B , denoted $A \cap B$, is the set of all elements that are in both A and B .
- The *difference* of A and B , denoted $A \setminus B$, is the set of all elements that are in A but not in B .
- The *complement* of A , denoted $\bar{A}$ or A^c , is the set of all elements in the universal set U that are not in A .

Definition (Intervals). An *interval* is a subset of the real line $\mathbb{R}$ that contains all real numbers between any two numbers in the set. Let $a, b \in \mathbb{R}$ with $a < b$.

- **Open Interval:** $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.
- **Closed Interval:** $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.
- **Half-Open Intervals:** $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$ and $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$.
- **Infinite Intervals:** $(a, \infty) = \{x \in \mathbb{R} \mid x > a\}$, $(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\}$.

Definition (Absolute Value). The *absolute value* of a real number x , denoted $|x|$, is defined as:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Properties of Absolute Value.

$$\begin{aligned} |ab| &= |a||b|, & \left|\frac{a}{b}\right| &= \frac{|a|}{|b|} \quad (b \neq 0), \\ |a + b| &\leq |a| + |b|, & |a - b| &\geq ||a| - |b||. \end{aligned}$$

Section 2 Functions and Their Graphs

Definition (Function). A *function* f from a set D to a set E is a rule that assigns to each element x in D exactly one element, called $f(x)$, in E . The set D is called the *domain* of f . The set E is called the *codomain* of f . The *range* of f is the set of all possible values $f(x)$ as x varies throughout the domain. We denote this mapping by $f : D \rightarrow E$. **Remark (Vertical Line Test).** A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once. **Definition (Piecewise Defined Functions).** A function may be defined by different formulas in different parts of its domain. Such a function is called a *piecewise defined function*.

Example 1 The absolute value function $f(x) = |x|$ is a piecewise defined function.

Definition (Even and Odd Functions). Let f be a function with domain D symmetric about the origin (i.e., if $x \in D$, then $-x \in D$).

- f is an *even function* if $f(-x) = f(x)$ for all $x \in D$. The graph of an even function is symmetric with respect to the y -axis.
- f is an *odd function* if $f(-x) = -f(x)$ for all $x \in D$. The graph of an odd function is symmetric with respect to the origin.

Definition (Increasing and Decreasing Functions). A function f is called:

- *increasing* on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .
- *decreasing* on an interval I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .

Section 3 Shifts and Scaling of Graphs

Given a basic function $y = f(x)$ and a constant $c > 0$, we can transform its graph in the following ways:

3.1 Vertical and Horizontal Shifts

- **Vertical Shift:** $y = f(x) + c$ shifts the graph of f **up** by c units. $y = f(x) - c$ shifts it **down** by c units.
- **Horizontal Shift:** $y = f(x - c)$ shifts the graph of f **right** by c units. $y = f(x + c)$ shifts it **left** by c units.

3.2 Scaling and Reflection

- **Vertical Scaling:** $y = cf(x)$ stretches the graph vertically by a factor of c if $c > 1$, and compresses it if $0 < c < 1$.
- **Horizontal Scaling:** $y = f(cx)$ compresses the graph horizontally by a factor of c if $c > 1$, and stretches it if $0 < c < 1$.
- **Reflections:**
 - $y = -f(x)$ reflects the graph across the x -axis.
 - $y = f(-x)$ reflects the graph across the y -axis.

Example 2 Consider $f(x) = x^2$. The graph of $y = (x - 2)^2 + 3$ is obtained by shifting the parabola $y = x^2$ right by 2 units and up by 3 units.

Section 4 Elementary and Transcendental Functions

Definition (Elementary Functions). Elementary functions are functions built from basic building blocks using algebraic operations (addition, subtraction, multiplication, division) and composition. These include:

- **Polynomials:** $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where $n \geq 0$ is an integer and a_i are constants. The domain is $\mathbb{R}$.
- **Rational Functions:** $f(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials. The domain is all x such that $Q(x) \neq 0$.

- **Power Functions:** $f(x) = x^a$, where a is a constant.
- **Root Functions:** $f(x) = \sqrt[n]{x}$.

Definition (Transcendental Functions). Transcendental functions are functions that are not algebraic. They cannot be expressed as a finite combination of algebraic operations.

- **Exponential Functions:** $f(x) = a^x$ (where $a > 0, a \neq 1$). The domain is $\mathbb{R}$, and the range is $(0, \infty)$. The most important base is Euler's number $e \approx 2.71828$, giving $f(x) = e^x$.
- **Logarithmic Functions:** $f(x) = \log_a x$. The domain is $(0, \infty)$, and the range is $\mathbb{R}$. The natural logarithm is $f(x) = \ln x = \log_e x$.
- **Trigonometric Functions:** $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$.

Properties of Exponential and Logarithmic Functions.

$$\begin{aligned} a^{x+y} &= a^x a^y, & \log_a(xy) &= \log_a x + \log_a y, \\ (a^x)^y &= a^{xy}, & \log_a(x^r) &= r \log_a x, \\ a^{\log_a x} &= x, & \log_a(a^x) &= x. \end{aligned}$$

Section 5 Inverse Functions

Definition (One-to-One Function). A function f is called *one-to-one* (or injective) if it never takes on the same value twice; that is, $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$. Equivalently, $f(x_1) = f(x_2) \implies x_1 = x_2$. **Remark (Horizontal Line Test).** A function is one-to-one if and only if no horizontal line intersects its graph more than once. **Definition (Inverse Function).** Let f be a one-to-one function with domain D and range R . Its *inverse function*, denoted f^{-1} , has domain R and range D , and is defined by:

$$f^{-1}(y) = x \iff f(x) = y.$$

Theorem (Properties of Inverse Functions). If f is a one-to-one function, then:

$$\begin{aligned} f^{-1}(f(x)) &= x \quad \text{for every } x \in D, \\ f(f^{-1}(y)) &= y \quad \text{for every } y \in R. \end{aligned}$$

Steps to Find the Inverse of a One-to-One Function.

1. Write $y = f(x)$.
2. Interchange x and y .
3. Solve this equation for y in terms of x .
4. The resulting expression is $y = f^{-1}(x)$.

Remark. The graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$.

Example 3 Find the inverse of $f(x) = 2x + 3$. Set $y = 2x + 3$. Interchange x and y : $x = 2y + 3$. Solve for y : $2y = x - 3 \implies y = \frac{x-3}{2}$. Thus, $f^{-1}(x) = \frac{x-3}{2}$.

Section 6 The Composition of Functions

Definition (Composition). Given two functions f and g , the *composite function* $f \circ g$ (read as “ f composed with g ”) is defined by:

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

Example 4 Let $f(x) = \sqrt{x}$ and $g(x) = x - 2$. Find $(f \circ g)(x)$ and its domain. We have $(f \circ g)(x) = f(g(x)) = f(x - 2) = \sqrt{x - 2}$. For this to be defined, we need $x - 2 \geq 0$, so $x \geq 2$. The domain is $[2, \infty)$.

Example 5 Let $f(x) = x^2$ and $g(x) = x + 1$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$$(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2.$$

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1.$$

Note that $f \circ g \neq g \circ f$ in general.

Section 7 Trigonometric Functions

Definition (Unit Circle Definition). Let t be a real number representing the length of an arc on the unit circle $x^2 + y^2 = 1$, starting from the point $(1, 0)$ and moving counterclockwise. If the terminal point of the arc is (x, y) , then:

$$\cos t = x, \quad \sin t = y, \quad \tan t = \frac{y}{x} \quad (x \neq 0).$$

Properties.

- **Periodicity:** $\sin(t + 2\pi) = \sin t$ and $\cos(t + 2\pi) = \cos t$. The sine and cosine functions have period 2π . The tangent function has period π .
- **Domain and Range:** The domain of $\sin t$ and $\cos t$ is $\mathbb{R}$, and their range is $[-1, 1]$. The domain of $\tan t$ is $t \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$.
- **Pythagorean Identity:** $\sin^2 t + \cos^2 t = 1$.
- **Sum and Difference Formulas:**

$$\sin(x + y) = \sin x \cos y + \cos x \sin y,$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y.$$

- **Double Angle Formulas:**

$$\sin(2x) = 2 \sin x \cos x,$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.$$

Graphs and Transformations. For functions of the form $y = A \sin(B(x - C)) + D$:

- A is the *amplitude* (vertical scaling). The graph oscillates between $D - |A|$ and $D + |A|$.
- B affects the *period*, where $\text{Period} = \frac{2\pi}{|B|}$.
- C is the *phase shift* (horizontal shift).

- D is the *vertical shift*.

Inverse Trigonometric Functions. The inverse functions of sine, cosine, and tangent are defined by restricting their domains to ensure they are one-to-one:

- $y = \arcsin x \iff \sin y = x$, for $x \in [-1, 1]$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.
- $y = \arccos x \iff \cos y = x$, for $x \in [-1, 1]$ and $y \in [0, \pi]$.
- $y = \arctan x \iff \tan y = x$, for $x \in \mathbb{R}$ and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$.